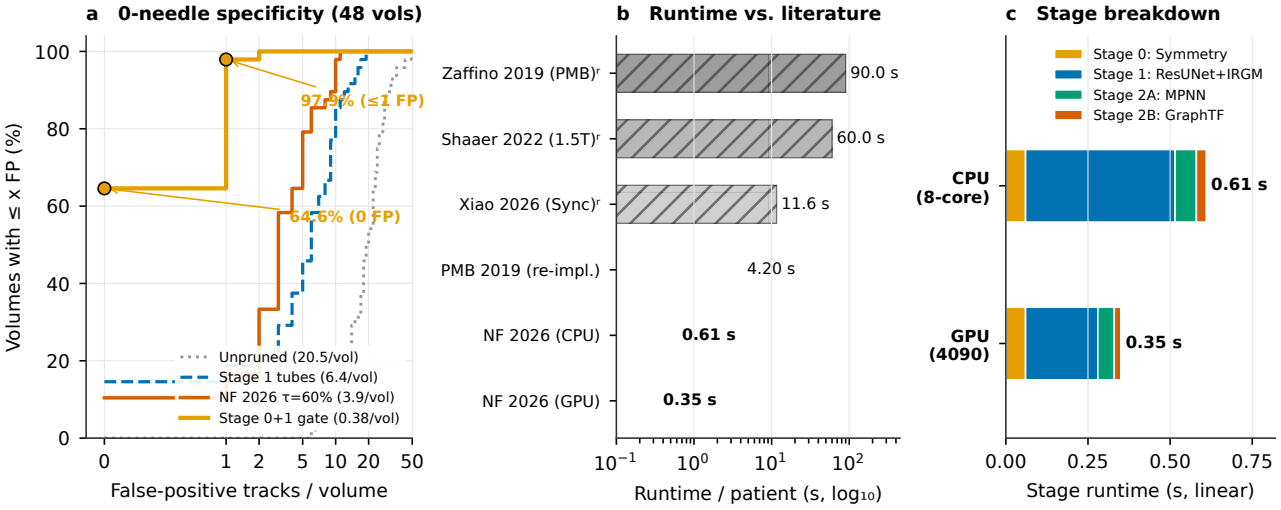
a 0-needle specificity (48 vols)
Volumes with ≤ x FP (%)
False-positive tracks / volume
97.9% (≤1 FP)
64.6% (0 FP)
Unpruned (20.5/vol)
Stage 1 tubes (6.4/vol)
NF 2026 τ=60% (3.9/vol)
Stage 0+1 gate (0.38/vol)
b Runtime vs. literature
Zaffino 2019 (PMB)ʳ
Shaaer 2022 (1.5T)ʳ
Xiao 2026 (Sync)ʳ
PMB 2019 (re-impl.)
NF 2026 (CPU)
NF 2026 (GPU)
90.0 s
60.0 s
11.6 s
4.20 s
0.61 s
0.35 s
Runtime / patient (s, log₁₀)
c Stage breakdown
Stage 0: Symmetry
Stage 1: ResUNet+IRGM
Stage 2A: MPNN
Stage 2B: GraphTF
CPU (8-core)
GPU (4090)
0.61 s
0.35 s
Stage runtime (s, linear)